

MaximEyes® FHIR API Documentation

Introduction

The MaximEyes FHIR APIs enable applications used by patients, clinicians, or health systems to connect and interact with MaximEyes by First Insight Corporation products. Certified APIs, as documented, are available in the current live versions of MaximEyes EHR version 3.0 (and above) and MaximEyes.com version 1.1 (and above). MaximEyes APIs use [HL7® FHIR® Release 4](#) for the exchange of [U.S. Core Data for Interoperability](#) version 3.1.1 data classes and elements.

This documentation is intended for use by third-party application developers creating products using FHIR specifications to exchange healthcare information electronically.

First Insight offers access to certified FHIR API endpoints free of charge for developers of apps used by patients and healthcare providers.

MaximEyes Developer Portal

To access MaximEyes APIs, a developer must create an account and register an app in the [MaximEyes Developer Portal](#). Information required for registration is dependent on the type of application and may include First Name, Last Name, Email Address, Username, Application Name, Company or Developer Name, Application Brief Description or Value Statement, App Type, Redirect & Launch URL(s) and Security or Privacy Policy information.

Once an application has been designed, built, and tested within your own preview environment you can request preview testing and production access via the options given for the registered application.

The process for obtaining Production access varies by API framework. First Insight may require a final verification of the app before granting production access which may include a review of the app's technical specifications, privacy & security information or agreements and a demo of the solution. If you are a developing an app used by patients to view their personal health information and only seeking access to certified APIs, then you may be issued a Production client ID without additional verification from First Insight after registering an application in the [MaximEyes Developer Portal](#). If you are not a Personal Health Record app developer, you can use the portal to request access to preview testing of Certified APIs.

First Insight provides support services reasonably necessary to enable the effective development, deployment, and use of certified API technology by API Information Sources and API Users in production environments.

Configuration and Conformance

- All requests to the API are accessed through https connection and must include the access token along with required parameters described for each search type. If the access token is missing, invalid, or expired the API will return an unauthorized response with proper message.
- The App MUST read and parse JSON responses.
- Assure that sensitive information (authentication secrets, authorization codes, tokens) are transmitted ONLY to authenticated servers, over TLS version 1.2 or higher.
- Bulk Export applications may need to be setup to handle longer lasting connections based on the data that is being exported.
- Locally hosted MaximEyes EHR version 3.0 (and above) requires the FHIR service to be installed on the customers server and TLS 1.2 will need to be enabled.
- Cloud hosted MaximEyes EHR version 3.0 (and above) and MaximEyes.com version 1.1 (and above) does not require any additional software.
- [Capability Statement](#) includes information about supported FHIR resources and security protocols and should be used by applications as the set of rules the application should follow.

Authorization

MaximEyes FHIR APIs use [SMART Application Launch 1.0.0](#), OAuth 2.0 and [Bulk Data 1.0.1](#) framework as methods through which an app requests authorization to access a FHIR resource, and then uses that authorization to retrieve the resource.

With a valid access token, an App can access data by issuing a FHIR API call to one or more of the FHIR endpoint(s) on the EHR's resource server. The request includes an Authorization header that presents the access_token as a "Bearer" token.

```
Example: Patient (ID 103007) using an access_token
GET https://FHIRDEHR.maximeyes.com/api/<CustomerName>/R4/Patient?qucryst=
patient3D103007
Authorization: Bearer IEzIny22OJPLHBhDKXqkSJeSbx5kwTC4MBmk9u8QFk8
```

Patient & Provider Apps Workflow

1. Call the Metadata API to get Token and Authorize endpoint @ <https://fhirdehr.maximeyes.com/api/FhirTest/R4/metadata>
2. App calls Authorize API with the below parameters.
At launch time, the app constructs a request for authorization by adding the following parameters to the query component of the EHR's "authorize" endpoint URL.:

Parameters

response_type	required	Fixed value: code.
client_id	required	The client's identifier.
redirect_uri	required	Must match one of the client's pre-registered redirect URIs.
launch	optional	When using the EHR launchflow, this must match the launch value received from the EHR.
scope	required	Must describe the access that the app needs, including clinical data scopes like patient/*.read, openid and fhirUser (if app needs authenticated patient identity) and either: - a launch value indicating that the app wants to receive already-established launch context details from the EHR - a set of launch context requirements in the form launch/patient, which asks the EHR to establish context on your behalf. See SMART on FHIR Access Scopes for details.
state	required	An opaque value used by the client to maintain state between the request and callback. The authorization server includes this value when redirecting the user-agent back to the client. The parameter SHALL be used for preventing cross-site request forgery or session fixation attacks.
aud	required	URL of the EHR resource server from which the app wishes to retrieve FHIR data. This parameter prevents leaking a genuine bearer token to a counterfeit resource server. (Note: in the case of an EHR launch flow, this aud value is the same as the launch's iss value.)

Example: [https://connect.first-insight.com/connect/authorize?response_type=code&client_id=patient&redirect_uri=https://inferno.healthit.gov/suites/custom/smart/redirect&scope=launch/patient openid fhirUser offline_access patient/Medication.read patient/AllergyIntolerance.read patient/CarePlan.read patient/CareTeam.read patient/Condition.read patient/Device.read patient/DiagnosticReport.read patient/DocumentReference.read patient/Encounter.read patient/Goal.read patient/Immunization.read patient/Location.read patient/MedicationRequest.read patient/Observation.read patient/Organization.read patient/Patient.read patient/Practitioner.read patient/Procedure.read patient/Provenance.read patient/PractitionerRole.read&state=67596367-b59a-46fc-bc07-b5bef5fe1a38&aud=https://fhirdehr.maximeyes.com/api/illmEr/R4](https://connect.first-insight.com/connect/authorize?response_type=code&client_id=patient&redirect_uri=https://inferno.healthit.gov/suites/custom/smart/redirect&scope=launch/patient%20openid%20fhirUser%20offline_access%20patient/Medication.read%20patient/AllergyIntolerance.read%20patient/CarePlan.read%20patient/CareTeam.read%20patient/Condition.read%20patient/Device.read%20patient/DiagnosticReport.read%20patient/DocumentReference.read%20patient/Encounter.read%20patient/Goal.read%20patient/Immunization.read%20patient/Location.read%20patient/MedicationRequest.read%20patient/Observation.read%20patient/Organization.read%20patient/Patient.read%20patient/Practitioner.read%20patient/Procedure.read%20patient/Provenance.read%20patient/PractitionerRole.read&state=67596367-b59a-46fc-bc07-b5bef5fe1a38&aud=https://fhirdehr.maximeyes.com/api/illmEr/R4)

3. App User is redirected to [MaximEyes Connect Authorization Server](#)

- If a Provider Login is entered the user will be redirected to patient selection screen. If Launch/encounter is in Scope, the Provider user will receive the encounter selection

screen. Following selections the user is redirected to the consent screen where they can authorize or deny access to resources.

- b. If a Patient Login is entered the user will be redirected to consent screen where they can authorize or deny access to resources.

4. App User is returned to the Client application (with authorization code).

Example: https://inferno.healthit.gov/suites/custom/smart/redirect?code=MNLWlJnl4vu0oHqkCCo_4qINg1Rpy-85eBFyJreh4R4&state=67596367-b59a-46fc-bc07-b5bef5fe1a38

5. Client App will call the Token API

After obtaining an authorization code, the app trades the code for an access token via HTTP POST to the EHR authorization server's token endpoint URL, using content-type application/x-www-form-urlencoded.

For **public apps**, authentication is not possible (and thus not required) since a client with no secret cannot prove its identity when it issues a call. (The end-to-end system can still be secure because the client comes from a known, https protected endpoint specified and enforced by the redirect uri.)

For **confidential apps**, an Authorization header using HTTP Basic authentication is required, where the username is the app's client_id and the password is the app's client_secret.

Parameters

grant_type	required	Fixed value: authorization_code
code	required	Code that the app received from the authorization server
redirect_uri	required	The same redirect_uri used in the initial authorization request
client_id	conditional	Required for public apps. Omit for confidential apps.

The EHR authorization server SHALL return a JSON object that includes an access token or a message indicating that the authorization request has been denied. The JSON structure includes the following parameters:

Parameters

access_token	required	The access token issued by the authorization server
token_type	required	Fixed value: Bearer
expires_in	recommended	Lifetime in seconds of the access token, after which the token SHALL NOT be accepted by the resource server
scope	required	Scope of access authorized. Note that this can be different from the scopes requested by the app.

id_token	optional	Authenticated patient identity and user details, if requested
refresh_token	optional	Token that can be used to obtain a new access token, using the same or a subset of the original authorization grants

Example:

POST

<https://connect.first-insight.com/connect/token>

code=MNLWlJnl4vu0oHqkCCo_4qINg1Rpy-

85eBFyJreh4R4&grant_type=authorization_code&redirect_uri=https%3A%2F%2Finferno.healthit.gov%2Fsuites%2Fcustom%2Fsmart%2Fredirect

10. Client App will get access_token to access the FHRI APIs

Standalone Launch

User selects an app from outside the EHR. This app will launch from its registered URL without a launch id.

App will call the Metadata API to get the Authorize and Token endpoints

And then app will call the Authorize endpoint with the requested scope

Example: [https://connect.first-](https://connect.first-insight.com/connect/authorize?response_type=code&client_id=patient&redirect_uri=https://inferno.healthit.gov/suites/custom/smart/redirect&scope=launch/patient openid fhirUser offline_access patient/Medication.read patient/AllergyIntolerance.read patient/CarePlan.read patient/CareTeam.read patient/Condition.read patient/Device.read patient/DiagnosticReport.read patient/DocumentReference.read patient/Encounter.read patient/Goal.read patient/Immunization.read patient/Location.read patient/MedicationRequest.read patient/Observation.read patient/Organization.read patient/Patient.read patient/Practitioner.read patient/Procedure.read patient/Provenance.read patient/PractitionerRole.read&state=67596367-b59a-46fc-bc07-b5bef5fe1a38&aud=https://fhirdehr.maximeyes.com/api/illmEr/R4)

[insight.com/connect/authorize?response_type=code&client_id=patient&redirect_uri=https://inferno.healthit.gov/suites/custom/smart/redirect&scope=launch/patient openid fhirUser offline_access patient/Medication.read patient/AllergyIntolerance.read patient/CarePlan.read patient/CareTeam.read patient/Condition.read patient/Device.read patient/DiagnosticReport.read patient/DocumentReference.read patient/Encounter.read patient/Goal.read patient/Immunization.read patient/Location.read patient/MedicationRequest.read patient/Observation.read patient/Organization.read patient/Patient.read patient/Practitioner.read patient/Procedure.read patient/Provenance.read patient/PractitionerRole.read&state=67596367-b59a-46fc-bc07-b5bef5fe1a38&aud=https://fhirdehr.maximeyes.com/api/illmEr/R4](https://connect.first-insight.com/connect/authorize?response_type=code&client_id=patient&redirect_uri=https://inferno.healthit.gov/suites/custom/smart/redirect&scope=launch/patient openid fhirUser offline_access patient/Medication.read patient/AllergyIntolerance.read patient/CarePlan.read patient/CareTeam.read patient/Condition.read patient/Device.read patient/DiagnosticReport.read patient/DocumentReference.read patient/Encounter.read patient/Goal.read patient/Immunization.read patient/Location.read patient/MedicationRequest.read patient/Observation.read patient/Organization.read patient/Patient.read patient/Practitioner.read patient/Procedure.read patient/Provenance.read patient/PractitionerRole.read&state=67596367-b59a-46fc-bc07-b5bef5fe1a38&aud=https://fhirdehr.maximeyes.com/api/illmEr/R4)

EHR Launch

In SMART's EHR launch flow, a user has established an EHR session, and then decides to launch an app. This could be a single-patient app or a user-level app. The EHR initiates a "launch sequence" by opening a new browser instance (or iframe) pointing to the app's registered launch URL and passing some context.

Parameters

iss	required	Identifies the EHR's FHIR endpoint, which the app can use to obtain additional details about the EHR, including its authorization URL.
-----	----------	--

launch	required	Opaque identifier for this specific launch, and any EHR context associated with it. This parameter must be communicated back to the EHR at authorization time by passing along a launch parameter (see example below).
--------	----------	--

Example:

EHR will open below URL in iframe

`https://<<app_launch_url>>?iss=<<fhir_endpoint_url>>&launch=xyz123`

Launch parameter as below

`{"pi":"10012","ei":"5214","ba":false}`

pi - Patient Number (So if this is passed then patient selection screen will be skipped and this patient get selected by default)

ei - Exam Number (If app want to open with specific encounter context and then with this it will bypass the encounter selection screen)

ba - Show banner, if this flag is send as false then it will return the need_patient_banner value as false in the token API response, the actual use of this will be based on this flag app will decide if they want to show patient banner in app, so in EHR launch if EHR already showing banner and opening a app in the iframe then if they set this to false app will not show banner so in this way the patient banner will nto be shown twice (one from EHR and one from App)

System Apps Workflow

1. Call the Token endpoint

Before a client can request an access token, it SHALL generate a one-time-use JSON Web Token (JWT) that will be used to authenticate the client to the FHIR authorization server. The authentication JWT SHALL include the following claims, and SHALL be signed with the client's private key (which SHOULD be an RS384 or ES384 signature). For a practical reference on JWT, as well as debugging tools and client libraries, see <https://jwt.io>.

Authentication JWT Header Values

alg	required	The JWA algorithm (e.g., RS384, ES384) used for signing the authentication JWT.
kid	required	The identifier of the key-pair used to sign this JWT. This identifier SHALL be unique within the client's JWK Set.
typ	required	Fixed value: JWT.
jku	optional	The TLS-protected URL to the JWK Set containing the public key(s) accessible without authentication or authorization. When present, this should match a value that the client supplied to the FHIR server at client registration time. (When absent, the FHIR server SHOULD fall back on the JWK Set URL or the JWK Set supplied at registration time.

Authentication JWT Claims

iss	required	Issuer of the JWT -- the client's client_id, as determined during registration with the FHIR authorization server (note that this is the same as the value for the sub claim)
sub	required	The service's client_id, as determined during registration with the FHIR authorization server (note that this is the same as the value for the iss claim)
aud	required	The FHIR authorization server's "token URL" (the same URL to which this authentication JWT will be posted -- see below)
exp	required	Expiration time integer for this authentication JWT, expressed in seconds since the "Epoch" (1970-01-01T00:00:00Z UTC). This time SHALL be no more than five minutes in the future.
jti	required	A nonce string value that uniquely identifies this authentication JWT.

After generating an authentication JWT, the client requests a new access token via HTTP POST to the FHIR authorization server's token endpoint URL, using content-type application/x-www-form-urlencoded with the following parameters:

Parameters

scope	required	The scope of access requested. See note about scopes below
grant_type	required	Fixed value: client_credentials
client_assertion_type	required	Fixed value: urn:ietf:params:oauth:client-assertion-type:jwt-bearer
client_assertion	required	Signed authentication JWT value (see above)

2. Response format**Access token response: property names**

access_token	required	The access token issued by the authorization server.
--------------	----------	--

token_type	required	Fixed value: bearer.
expires_in	required	The lifetime in seconds of the access token. The recommended value is 300, for a five-minute token lifetime.
scope	required	Scope of access authorized. Note that this can be different from the scopes requested by the app.

Example:

POST

<https://connect.first-insight.com/connect/token>

client_assertion=eyJ0eXAiOiJKV1QiLCJhbGciOiJFUzM4NCIsImtpZCI6IjRiNDIhNzU5ZDFiYjExNWVzZmJlZjRjZjliZWl2ZDFiLn0.eyJpc3MiOiJidWxrZXhwb3J0Iiwic3ViljoiYnVsa2V4cG9ydCIsImF1ZCI6Imh0dHBzOi8vY29ubmVjdC5maXJzdC1pbmNpZ2h0LmNvbS9jb25uZWNOIiwiaWF0IjE5MjQ5NTAxZTI3ZWl2ZjNjYzNjc1MjA0LCJqdGkiOiJOTYzNTk2OWE0ODg5MDc3MDkzZTNmNzNiY2Q5NTAxZTI3ZWl2ZjM0ODBmN2ZmMjc5OTY0YzRmNWUzMmEwY2FhLn0.h6tBCKXqZcxRgl_mrL5Sw7VmeUvfA7ld_BdjQJE62hiBpPp0NFtxGLsigkx1gLpxSpo51dhPERjScJA_LWSnhWGYBrM4fhI01wp-4CqgEeCWBpR_eNKPjCNii_0Ezis0&client_assertion_type=urn%3Aietf%3Aparams%3Aoauth%3Aclient-assertion-type%3Ajwt-bearer&grant_type=client_credentials&scope=system%2F%2A.read

Response:

```
{
  "access_token": "jJAdg1D41qJBfdfa_YJJpnmiDgBadv1hB_cosjhVkn0",
  "token_type": "Bearer",
  "expires_in": 3599,
  "scope": "system/*.read"
}
```

- With a valid access token, a client MAY issue a FHIR API call to a FHIR resource server

FHIR Base URLs

The base URLs for our supported environments are as seen below.

<Base-URL>: product.maximeyes.com

MaximEyes EHR	https://FHIRDEHR.maximeyes.com
MaximEyes.com	https://fhirwehr.maximeyes.com

MaximEyes FHIR API swagger URL details are seen as below.

MaximEyes EHR	https://FHIRDEHR.maximeyes.com/api/FHIR/R4
MaximEyes.com	https://fhirwehr.maximeyes.com/api/FHIR/R4

Bulk Export

[Bulk Data 1.0.1](#) is used by developers of backend services to access resources from FHIR servers that have pre-authorized defined scopes of access.

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Export Status API Search Parameters

Search Parameter	Value	Example
GUID	Guid Value	Get[base]/CustomerName/r4/status?Guid=<Value

Delete API Search Parameters

Search Parameter	Value	Example
GUID	Guid Value	DELETE [base]/CustomerName/r4/status?Guid=<Value

Supported Scopes

Patient Scopes:

- patient/*.read
- patient/AllergyIntolerance.read
- patient/CarePlan.read
- patient/CareTeam.read
- patient/Condition.read
- patient/Device.read
- patient/DiagnosticReport.read
- patient/DocumentReference.read
- patient/Encounter.read
- patient/Goal.read
- patient/Immunization.read
- patient/Location.read
- patient/Medication.read
- patient/MedicationRequest.read
- patient/Observation.read
- patient/Organization.read
- patient/Patient.read
- patient/Practitioner.read
- patient/PractitionerRole.read
- patient/Procedure.read
- patient/Provenance.read

User Scopes:

- user/*.read
- user/AllergyIntolerance.read
- user/CarePlan.read
- user/CareTeam.read
- user/Condition.read
- user/Device.read
- user/DiagnosticReport.read
- user/DocumentReference.read
- user/Encounter.read
- user/Goal.read
- user/Immunization.read
- user/Location.read
- user/Medication.read
- user/MedicationRequest.read
- user/Observation.read
- user/Organization.read
- user/Patient.read
- user/Practitioner.read
- user/PractitionerRole.read
- user/Procedure.read

- user/Provenance.read

System

- system/*.read
- system/AllergyIntolerance.read
- system/CarePlan.read
- system/CareTeam.read
- system/Condition.read
- system/Device.read
- system/DiagnosticReport.read
- system/DocumentReference.read
- system/Encounter.read
- system/Goal.read
- system/Immunization.read
- system/Location.read
- system/Medication.read
- system/MedicationRequest.read
- system/Observation.read
- system/Organization.read
- system/Patient.read
- system/Practitioner.read
- system/PractitionerRole.read
- system/Procedure.read
- system/Provenance.read

Additional Supported Scopes

- launch/patient (When launching outside the EHR, ask for a patient to be selected at launch time)
- launch/encounter (When launching outside the EHR, ask for an encounter to be selected at launch time)
- Launch (Permission to obtain launch context when app is launched from an EHR)
- Openid fhirUser (Permission to retrieve information about the current logged-in user)
- Offline_access (Request a refresh_token that can be used to obtain a new access token to replace an expired one, even after the end-user no longer is online after the access token expires)
- Online_access (Request a refresh_token that can be used to obtain a new access token to replace an expired one, and that will be usable for as long as the end-user remains online.)

Proof Key for Code Exchange (PKCE)

MaximEyes FHIR APIs support S256 code_challenge_method for all apps. PKCE is enabled automatically for all apps during registration. PKCE is required for public apps, but it can be disabled for confidential apps if specifically requested by the app developer.

When PKCE is in the authorization request, the client needs to send 2 extra parameters:

code_challenge	required	This parameter is generated by the app and used for the code challenge, as specified by PKCE . For example, when code_challenge_method is 'S256', this is the S256 hashed version of the code_verifier parameter. See considerations-for-pkce-support .
code_challenge_method	required	Method used for the code_challenge parameter. Example value: S256. See considerations-for-pkce-support .

Client needs to pass the code_verifier extra parameter to token request.

code_verifier	required	This parameter is used to verify against the code_challenge parameter previously provided in the authorize request.
---------------	----------	---

FHIR R4 APIs

Action	Method	URL
Show	GET	<a href="https://<<practicePublicIP>>/api/<<urlsubdomain>>/R4">https://<<practicePublicIP>>/api/<<urlsubdomain>>/R4 e.g. For the product MaximEyes EHR https://FHIRDEHR.maximeyes.com/api/FHIR/R4 For the product MaximEyes.com https://fhirwehr.maximeyes.com/api/FHIR/R4
Search	GET	<a href="https://<<practicePublicIP>>/api/<<urlsubdomain>>/R4">https://<<practicePublicIP>>/api/<<urlsubdomain>>/R4 e.g. For the product MaximEyes EHR https://FHIRDEHR.maximeyes.com/api/FHIR/R4 For the product MaximEyes.com https://fhirwehr.maximeyes.com/api/FHIR/R4

AllergyIntolerance

The [AllergyIntolerance](#) resource returns data classes and elements defined for Allergies and Intolerances in [USCDI](#).

- Substance (Medication)
- Substance (Drug Class)
- Reaction

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	_id	PT_ENCOUNTER_ALLERGY_HISTORY_RECORD_ID	GET [base]/AllergyIntolerance?_id=[id] E.g. AllergyIntolerance?_id=123
SHALL	patient	Patient Number	GET [base]/AllergyIntolerance?patient=[id] E.g.1) AllergyIntolerance?patient=12345 2) AllergyIntolerance?patient=Patient/12345
SHALL	patient+clinical-status	Patient Number & Status (Which is selected at the time of allergy created)	GET [base]/AllergyIntolerance?patient=[id]&clinical-status=[status] E.g. AllergyIntolerance?patient=12345&clinical-status=Active

Request Parameters

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/AllergyIntolerance?[parameter=value]&_revinclude=Provenance:target

CarePlan

The [CarePlan](#) resource returns data classes and elements defined for Assessment and Plan of Treatment in [USCDI](#).

- Assessment and Plan of Treatment

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	patient+category	Patient ID & Category=assess-plan (assess-plan is fixed value for category)	GET [base]/CarePlan?patient=<PatientNumber>&category=assess-plan
SHOULD	patient+category+date	Patient ID & Category=assess-plan (assess-plan is fixed value for category) & date=(ENCOUNTER_DATE)	GET [base]/CarePlan?patient=<PatientNumber>&category=assess-plan&date=<EncounterDate>
SHOULD	patient+category+status,	Patient ID & Category=assess-plan&Status=Active (Note :Category = Access-Plan and Status= Active will be fixed values)	GET [base]/CarePlan?patient=<PatientNumber>&category=assess-plan&status=active
SHOULD	patient+category+date ,	Patient ID & Category=assess-plan&Status=Active&date=(Encounter_Date) (Note :Category = Access-Plan and Status= Active will be fixed values)	GET [base]/CarePlan?patient=<PatientNumber>&category=assess-plan&status=active&date=<EncounterDate>
		Patient ID	POST [base]/CarePlan?patient=PatientNumber>

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/CarePlan? category=assess-plan&patient=[parameter=value]&_revinclude=Provenance:target

CareTeam

The [CareTeam](#) resource returns data classes and elements defined for Care Team Member(s) in [USCDI](#).

- Care Team

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	patient+status	Patient ID &Status=Active Note : Status =Active will be a fixed value(It is fixed)	GET [base]/CareTeam?patient=<PatientNumber>&status=active
		Patient ID	POST [base]/CareTeam?patient=<PatientNumber>

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/ CareTeam?parameter=value&status=active&_revinclude=Provenance:target

Condition

The [Condition](#) resource returns data classes and elements defined for Health Concerns in [USCDI](#).

- Health Concerns
- Problems

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	patient	Patient ID	GET [base]/Condition?patient=<Patient Number>

Conformance	Search Parameter	Value	Example
SHOULD	patient+status	Patient ID &Status=Active Note: Status =Active will be a fixed value(It is fixed)	GET [base]/CareTeam?patient=<PatientNumber>&status=active
SHOULD	Patient+category	Patient ID & Category	GET [base]/Condition?patient=<PatientNumber>&category=http://terminology.hl7.org/CodeSystem/condition-category problem-list-item GET [base]/Condition?patient=<PatientNumber>&category=http://hl7.org/fhir/us/core/CodeSystem/condition-category health-concern GET [base]/Condition?patient=<PatientNumber>&category=http://terminology.hl7.org/CodeSystem/condition-category encounter-diagnosis
SHOULD	Patient+code	Patient ID & code	GET [base]/Condition?patient=<PatientNumber>&code=[http://snomed.info/sct 442311008]
SHOULD	Patient+onset-date	Patient ID & date	GET [base]Condition?patient=<PatientNumber>&onset-date={gt lt ge le}[date]&onset-date={gt lt ge le}[date]&...}
		Patient ID	POST [base]/CareTeam?patient=<PatientNumber>

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/ Condition?[parameter=value]&_revinclude=Provenance:target

Implantable Device

The [Implantable Device](#) resource returns data classes and elements defined for Unique Device Identifier(s) for a Patient's Implantable Device(s) in USCDI.

- Unique Device Identifier(s) for Patients' Implantable Device(s)

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	patient	Patient ID	GET [base]/Device?patient=<PatientNumber>
SHOULD	patient+type	Patient ID & type=(PT_EN COUNTER_I MPLANTED_ DEVICES --> SNOMED_CO DE)	GET [base]/Device?patient=<PatientNumber> &type=468063009

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/ Device?[parameter=value]&_revinclude=Provenance:target

DiagnosticReport

The [DiagnosticReport](#) resource returns data classes and elements defined for Laboratory in [USCDI](#).

- Imaging Narrative
- Laboratory Report Narrative
- Pathology Report Narrative
- Procedure Note

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	Patient	Patient ID	GET [base]/DiagnosticReport?patient=<PatientNumber>
SHALL	Patient+category	Patient ID & category=Lab Note: Category value is fixed	GET [base]/DiagnosticReport?patient=<PatientNumber>&category=LAB
SHALL	Patient+code	PatientID&Code	GET [base]/DiagnosticReport?patient=<PatientNumber>&code=24323-8
SHALL	Patient+category+date	Patient ID & category=Lab & date=(TEST_ORDERED_DATE)	GET [base]/DiagnosticReport?patient=<PatientNumber>&category=LAB&date={gt lt ge le}[date]&date={gt lt ge le}[date]&...}
SHOULD	Patient + status	patient id & status=Final Note: status value is fixed.	GET [base]/DiagnosticReport?patient=1137192&status=completed
SHOULD	Patient+code+date	Patient ID & code=(Loinc_code) & date=(TEST_ORDERED_DATE)	GET [base]/DiagnosticReport?patient=<PatientNumber>&code=24323-8&date={gt lt ge le}[date]&date={gt lt ge le}[date]&...}
		Patient ID & category=LAB	POST [base]/DiagnosticReport?patient=<PatientNumber>&category=LAB

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/ DiagnosticReport?[parameter=value]&_revinclude=Provenance:target

DocumentReference

The [DocumentReference](#) resource returns data classes and elements defined for Clinical Notes in [USCDI](#).

- Consultation Note
- Discharge Summary Note
- History & Physical
- Progress Note

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	_id	PATIENT_ENCOUNTER_ID	GET [base]/DocumentReference[id]
SHALL	patient	Patient Number	GET [base]/DocumentReference?patient={Type/}[id]
SHALL	Patient + type	Patient Number + type	GET [base]/DocumentReference?patient={Type/}[id]&type={system }[code]
SHALL	patient+category+date	Patient Number + category=clinical-note + date (ENCOUNTER_DATE)	GET [base]/DocumentReference?patient={Type/}[id]&category=clinical-note&date=
SHALL	patient+category	Patient Number + category=clinical-note	GET [base]/DocumentReference?patient={Type/}[id]&category=clinical-note
SHOULD	patient+status	Patient Number + status	GET [base]/DocumentReference?patient={Type/}[id]&status=Current

Conformance	Search Parameter	Value	Example
SHOULD	patient+type+period	Patient Number + type + date	GET [base]/DocumentReference?patient=[reference]&type={system }[code]&period={gt lt ge le}[date]{&period={gt lt ge le}[date]&...}

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/DocumentReference?[parameter=value]&_revinclude=Provenance:target

Encounter

The [Encounter](#) resource returns data classes and elements defined for Encounter Information in [USCDI](#).

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	_id	EXAM_NUMBER	GET [base]/Encounter/<ExamNumber>
SHALL	patient	Patient Number	GET [base]/Encounter?patient=<PatientNumber>
SHALL	date+patient	Patient ID & ENCOUNTER_DATE	GET [base]/Encounter?patient=<PatientNumber>&date={gt lt ge le}[date]{&date={gt lt ge le}[date]&...}
SHOULD	identifier	Patient Number	GET [base]/Encounter?identifier=<PatientNumber>
SHOULD	class+patient	Patient Number&Class=AMB Note: AMB will be the fixed values	GET [base]/Encounter?patient=<PatientNumber>&class=AMB

Conformance	Search Parameter	Value	Example
SHOULD	Patient+type	Patient Number & type	GET [base]/Encounter?patient=[reference]&type={system }[code]
SHOULD	Patient+status	Patient Number & Status=Finished Note: Finished will be the fixed value	GET [base]/Encounter?patient=[reference]&status={system }[code]

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Encounter?[parameter=value]&_revinclude=Provenance:target

Goal

The [Goal](#) resource returns data classes and elements defined for Goals in [USCDI](#).

- Patient Goals

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	patient	Patient ID	GET [base]/Goal?patient=<PatientNumber>
SHOULD	patient+target-date	Patient ID&TargetDate	GET [base]/Goal?patient=<PatientNumber>&target-date
SHOULD	patient+lifecycle-status	Patient ID & lifecycle-status=Active Note : Active is the fixed value	GET [base]/Goal?patient=<PatientNumber>&lifecycle-status=active
		Patient ID	POST [base]/DiagnosticReport?patient=135003

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/ Goal?[parameter=value]&_revinclude=Provenance:target

Immunization

The [Immunization](#) resource returns data classes and elements defined for Immunizations in [USCDI](#).

- Immunization

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Seach Parameter	Value	Example
SHALL	patient	Patient ID	GET [base]/Immunization? patient=<PatientNumb er>
SHOULD	patient+status	Patient ID & status=Completed Note : Completed is the fixed value for status)	GET [base]/Immunization? patient=<PatientNumb er>&status=completed
SHOULD	patient+date	patient id & date=(Table: PT_ENCOUNTER_IMM NUNIZATION_RECORD S --> DATE_ADMINISTERED)	GET [base]/Immunization? patient=<PatientNumb er>&date=date
		Patient ID	POST [base]/Immunization? patient=<PatientNumb er>

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Immunization?[parameter=value]&_revinclude=Provenance:target

Location

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token for [Location](#). Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	Name	Location Name	GET [base]/Location?name=[string]
SHALL	address	Address	GET [base]/Location?address=[string]
SHOULD	address-state	State	GET [base]/Location?address-city=[string]
SHOULD	address-city	City	GET [base]/Location?address-city=[string]
SHOULD	address-postalCode	postalcode	GET [base]/Location?address-postalcode=[string]

Medication Request

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token for [MedicationRequest](#). Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	patient+intent,	Patient ID & intent=order	GET [base]/MedicationRequest?patient=<Patient Number>&intent=order GET [base]/MedicationRequest?patient=<Patient Number>&&intent=order&_include=MedicationRequest:medication
SHALL	patient+intent+status,	Patient ID & intent=order & status=active	GET [base]/MedicationRequest?patient=<Patient Number>&&intent=order

Conformance	Search Parameter	Value	Example
			r&status=active GET [base]/MedicationRequest?patient=<PatientNumber>&intent=order&status=active&_include=MedicationRequest:medication
SHOULD	patient+intent+encounter	Patient ID & intent=order & encounter=Exam_number	GET [base]/MedicationRequest?patient=<PatientNumber>&intent=order&status=active&encounter=Encounter/123 GET [base]/MedicationRequest?patient=<PatientNumber>&intent=order&status=active&encounter=Encounter/123&_include=MedicationRequest:medication
SHOULD	patient+intent+authoredon	Patient ID & intent=order & authoredon=date(Table: PT_ENCOUNTER_MEDICATION_HISTORY_RECORDS --> date)	GET [base]/MedicationRequest?patient=<PatientNumber>&intent=order&authoredon=<Medication Date> GET [base]/MedicationRequest?patient=<PatientNumber>&intent=order&authoredon=<Medication Date>&_include=MedicationRequest:medication
		Patient ID & intent=order	POST [base]/MedicationRequest?patient=<PatientNumber>&intent=order

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/MedicationRequest?[parameter=value]&_revinclude=Provenance:target

Observation

The Observation resource returns data classes and elements defined for Laboratory in [USCDI](#).

- Tests
- Values/Results

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	Patient + category	patient id & category=Laboratory (Note: category value is fixed)	GET [base]/Observation?patient=[reference]&category=http://terminology.hl7.org/CodeSystem/observation-category laboratory
SHALL	Patient+code	patient id & code=(PT_ENCOUNTER_TEST_ORDERS-->LOINC_CODE)	GET [base]/Observation?patient=[reference]&code={system }{code }{system }{code },...
SHALL	Patient+category+date	patient id & category=Laboratory& date (Note: category value is fixed)	GET [base]/Observation?patient=[reference]&category=http://terminology.hl7.org/CodeSystem/observation-category laboratory&date={gt lt ge le}[date]&date={gt lt ge le}[date]&...
SHOULD	Patient+category+status	patient id & category=Laboratory & status=Final	GET [base]/Observation?patient=[reference]&category=http://terminology.hl7.org/CodeSystem

Conformance	Search Parameter	Value	Example
		Note: status value is fixed.	/observation-category laboratory&status={system }[code]{,system }[code],...}
SHOULD	Patient+code+date	Gender(Whatever it will be) + First Name	GET [base]/Observation?patient=[reference]&code={system }[code]{,system }[code],...}&date={gt lt ge le}[date]{&date={gt lt ge le}[date]&...}

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Smoking Status Observation

The [Smoking status Observation](#) resource returns data classes and elements defined for Smoking Status in [USCDI](#).

- Smoking Status

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	Patient+Code	patient id & Code	GET [base]/Observation?patient=<PatientNumber>&code=http://loinc.org 72166-2

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Vital Signs

Pediatric BMI for Age

The [Pediatric BMI for Age Observation](#) resource returns data classes and elements defined for Vital Signs in [USCDI](#).

- BMI Percentile (2 - 20 years)

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Search Parameters	Value	Example
Patient+Code	Note: code value is fixed for 'Pediatric BMI for Age Observation Tests'	GET [base]/Observation?patient={Type}/{id}&code={system }[code]
Patient+ Category+date	date = encounter date	GET [base]/Observation?patient={Type}/{id}&category=vital-signs&date={gt lt ge le}[date]
Patient+Code+Date		GET [base]/Observation?patient={Type}/{id}&code={system }[code]&date={gt lt ge le}[date]
Patient+Category+Status	Note: status=final (This is fixed)	GET [base]/Observation?patient={Type}/{id}&category=vital-signs&status={system }[code]
Patient+Code	Note: code value is fixed for 'Pediatric BMI for Age Observation Tests'	GET [base]/Observation?patient={Type}/{id}&code={system }[code]
Patient+ Category+date	date = encounter date	GET [base]/Observation?patient={Type}/{id}&category=vital-signs&date={gt lt ge le}[date]
Patient+code+Date		GET [base]/Observation?patient={Type}/{id}&code={system }[code]&date={gt lt ge le}[date]
Patient+Category+Status	Note: status=final (This is fixed)	GET [base]/Observation?patient={Type}/{id}&category=vital-signs&status={system }[code]

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Pediatric Weight for Height

The [Pediatric Weight for Height Observation](#) resource returns data classes and elements defined for Vital Signs in [USCDI](#).

- Weight-for-length Percentile

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Search Parameters	Value	Example
Patient+ Code	Note: code value is fixed for 'Pediatric Weight for Height Observation'	GET [base]/Observation?patient={Type}/{id}&code={system }[code]
Patient+Category+Date	date = encounter date	GET [base]/Observation?patient={Type}/{id}&category=vital-signs&date={gt lt ge le}[date]
Patient+Code+Date		GET [base]/Observation?patient={Type}/{id}&code={system }[code]&date={gt lt ge le}[date]
Patient+Category+Status	Note: status=final (This is fixed)	GET [base]/Observation?patient={Type}/{id}&category=vital-signs&status={system }[code]
Patient+Code	Note: code value is fixed for Smoking-Status	GET [base]/Observation?patient={Type}/{id}&code=http://loinc.org 72166-2
Patient+code	Note: code value is fixed for 'Pediatric Weight for Height Observation'	GET [base]/Observation?patient={Type}/{id}&code={system }[code]

Patient+Category+Date	date = encounter date	GET [base]/Observation?patient={Type}/{id}&category=vital-signs&date={gt lt ge le}[date]
-----------------------	-----------------------	---

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Pulse Oximetry

The [Pulse Oximetry](#) resource returns data classes and elements defined for Vital Signs in [USCDI](#).

- Pulse oximetry

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Search Parameters	Value	Example
Patient+Code	Note: code value is fixed for 'Pulse Oximetry Tests'	GET [base]/Observation?patient={Type}/{id}&code={system }[code]
Patient+ Category+date	date = encounter date	GET [base]/Observation?patient={Type}/{id}&category=vital-signs&date={gt lt ge le}[date]
Patient+code+date		GET [base]/Observation?patient={Type}/{id}&code={system }[code]&date={gt lt ge le}[date]
Patient+Category+status	Note: status=final (This is fixed)	GET [base]/Observation?patient={Type}/{id}&category=vital-signs&status={system }[code]
Patient+code	Note: code value is fixed for 'Pulse Oximetry Tests'	GET [base]/Observation?patient={Type}/{id}&code={system }[code]
Patient+ Category+date	date = encounter date	GET [base]/Observation?patient={T

Search Parameters	Value	Example
		type/}[id]&category=vital-signs&date={gt lt ge le}[date]
Patient+Code+date		GET [base]/Observation?patient={Type/}[id]&code={system }[code]&date={gt lt ge le}[date]
Patient+Category+Status	Note: status=final (This is fixed)	GET [base]/Observation?patient={Type/}[id]&category=vital-signs&status={system }[code]

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Occipital-frontal Circumference Percentile

The [Pediatric Head Occipital Frontal Circumference Observation](#) resource returns data classes and elements defined for Vital Signs in [USCDI](#).

- Head Occipital-frontal Circumference Percentile (Birth - 36 Months)

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Search Parameter	Value	Example
Patient+Code	Note: code value is fixed for 'Pediatric Head Occipital-frontal Circumference Percentile Tests'	GET [base]/Observation?patient=1186747&code=8289-1
Patient+ Category+date	date = encounter date	GET [base]/Observation?patient=1134281&category=vital-signs&date=ge2020-01-01
Patient+code+date		GET [base]/Observation?patient=1186747&code=8289-1&date=ge2020-01-01
Patient+Category+Status	Note: status=final (This is fixed)	GET [base]/Observation?patient=11

		34281&category=vital-signs&status=final
--	--	---

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Body Height

The Body height resource returns data classes and elements defined for Vital Signs in [USCDI](#).

- Body height

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Search Parameter	Value	Example
Patient+Code	Note: code value is fixed for 'Observation Body Height Tests'	GET [base]/Observation?patient=1186747&code=8302-2
Patient+ Category+date	date = encounter date	GET [base]/Observation?patient=1134281&category=vital-signs&date=ge2020-01-01
Patient+code+date		GET [base]/Observation?patient=1186747&code=8302-2&date=ge2020-01-01
Patient+Category+Status	Note: status=final (This is fixed)	GET [base]/Observation?patient=1134281&category=vital-signs&status=final

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Body Temperature

The Body temperature resource returns data classes and elements defined for Vital Signs in [USCDI](#).

- Body temperature

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Search Parameter	Value	Example
Patient+Code	Note: code value is fixed for 'Observation Body Temperature Tests'	GET [base]/Observation?patient=1186747&code=8310-5
Patient+ Category+date	date = encounter date	GET [base]/Observation?patient=1134281&category=vital-signs&date=ge2020-01-01
Patient+code+date		GET [base]/Observation?patient=1186747&code=8310-5&date=ge2020-01-01
Patient+Category+Status	Note: status=final (This is fixed)	GET [base]/Observation?patient=1134281&category=vital-signs&status=final

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Blood Pressure

The Blood pressure systolic and diastolic resource returns data classes and elements defined for Vital Signs in [USCDI](#).

- Systolic blood pressure
- Diastolic blood pressure

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Search Parameter	Value	Example
Patient+Code	Note: code value is fixed for 'Observation Blood Pressure Tests'	GET [base]/Observation?patient=1186747&code=85354-9
Patient+ Category+date	date = encounter date	GET [base]/Observation?patient=1134281&category=vital-signs&date=ge2020-01-01
Patient+code+date		GET [base]/Observation?patient=1186747&code=85354-9&date=ge2020-01-01
Patient+Category+Status	Note: status=final (This is fixed)	GET [base]/Observation?patient=1134281&category=vital-signs&status=final

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Body Weight

The Body weight resource returns data classes and elements defined for Vital Signs in [USCDI](#).

- Body weight

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Search Parameter	Value	Example
Patient+Code	Note: code value is fixed for 'Observation Blood Pressure Tests'	GET [base]/Observation?patient=1186747&code=85354-9
Patient+ Category+date	date = encounter date	GET [base]/Observation?patient=1134281&category=vital-signs&date=ge2020-01-01
Patient+code+date		GET [base]/Observation?patient=11

Search Parameter	Value	Example
		86747&code=85354-9&date=ge2020-01-01
Patient+Category+Status	Note: status=final (This is fixed)	GET [base]/Observation?patient=1134281&category=vital-signs&status=final

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Heart Rate

The Heart rate resource returns data classes and elements defined for Vital Signs in [USCDI](#).

- Heart Rate

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Search Parameter	Value	Example
Patient+Code	Note: code value is fixed for 'Observation Heart Rate Tests'	GET [base]/Observation?patient=1186747&code=8867-4
Patient+ Category+date	date = encounter date	GET [base]/Observation?patient=1134281&category=vital-signs&date=ge2020-01-01
Patient+code+date		GET [base]/Observation?patient=1186747&code=8867-4&date=ge2020-01-01
Patient+Category+Status	Note: status=final (This is fixed)	GET [base]/Observation?patient=1134281&category=vital-signs&status=final

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Respiratory Rate

The Respiratory rate resource returns data classes and elements defined for Vital Signs in [USCDI](#).

- Respiratory rate

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Search Parameter	Value	Example
Patient+Code	Note: code value is fixed for 'Observation Heart Rate Tests'	GET [base]/Observation?patient=1186747&code=9279-1
Patient+ Category+date	date = encounter date	GET [base]/Observation?patient=1134281&category=vital-signs&date=ge2020-01-01
Patient+code+date		GET [base]/Observation?patient=1186747&code=9279-1&date=ge2020-01-01
Patient+Category+Status	Note: status=final (This is fixed)	GET [base]/Observation?patient=1134281&category=vital-signs&status=final

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Observation?[parameter=value]&_revinclude=Provenance:target

Organization

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token for [Organization](#). Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
-------------	------------------	-------	---------

SHALL	Name	Name	GET [base]/Organization?name=[string]
SHALL	address	Address	GET [base]/Organization?address=[string]

Patient

The [Patient](#) resource returns data classes and elements defined for Patient Demographics/Information in [USCDI](#).

- First Name
- Middle Name
- Last Name
- Previous Name
- Suffix
- Birth Sex
- Date of Birth
- Race
- Ethnicity
- Preferred Language
- Address
- Phone Number

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	_id	Patient Number	GET [base]/Patient?_id=[id]
SHALL	identifier	Patient Number	GET [base]/Patient?identifier=[system] [code]
SHALL	name	Patient First Name or Last Name	GET [base]/Patient?name=[name]
SHALL	birthdate+name	DOB of Patient + Name of Patient (First name or Last name)	GET [base]/Patient?birthdate=[birthdate]&name=[name]
SHALL	gender+name	Gender(Whatever it will be) + First Name	GET [base]/Patient?gender=[system] [code]&name=[name]
SHOULD	birthdate+family	DOB of Patient + Last Name	GET [base]/Patient?birthdate=[date]&family=[string]
SHOULD	family+gender	Last Name + Gender(Whatever it will be)	GET [base]/Patient?family=[string]&gender={system} [code]

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Patient?[parameter=value]&_revinclude=Provenance:target

Practitioner

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token for [Practitioner](#). Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	Name	Name	GET [base]/Practitioner?name=[string]
SHALL	identifier	identifier	GET [base]/Practitioner?identifier={system }[code]

PractitionerRole

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token for [PractitionerRole](#). Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameter	Value	Example
SHALL	specialty	Specialty code	GET [base]/PractitionerRole?specialty={{system}} {code}&_include=PractitionerRole:practitioner}&_include=PractitionerRole?endpoint}
SHALL	practitioner	practitioner	GET [base]/PractitionerRole?practitioner=[reference]&_include=PractitionerRole:practitioner}&_include=PractitionerRole?endpoint}

Procedure

The [Procedure](#) resource returns data classes and elements defined for Procedures in [USCDI](#).

- Procedures

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameters	Value	Example
SHALL	patient	Patient ID	GET [base]/Procedure?patient=<PatientNumber>
SHALL	patient+date	Patient ID & code=(snomed code against category and type)& date=(Test_performed_date or encounter_date)	GET [base]/Procedure?patient=<PatientNumber>&date=<Test_performed_date or encounter_date >
SHOULD	patient+status	Patient ID & status	GET [base]/Procedure?patient=<PatientNumber>&status=<CurrentStatus>
SHOULD	patient+code+date	Patient ID & date=(TEST_PERFORMED_DATE or Encounter_date)	GET [base]/Procedure?patient=<PatientNumber>&date==<Test_performed_date or encounter_date >
		Patient ID	POST [base]/Procedure?patient=<PatientNumber>

Request Parameter

The “_revinclude” search result parameter with any request will include additional resources, based on following reverse links across references.

GET [base]/Procedure?[parameter=value]&_revinclude=Provenance:target

Provenance

The [Provenance](#) resource returns data classes and elements defined for Provenance in [USCDI](#).

- Author Time Stamp
- Author Organization

Data response and return examples can be viewed at [MaximEyes FHIR API Swagger](#) with or without an access token. Examples are given without an access token or parameter entries. Authorize an access token to retrieve responses based on parameter entries.

Search Parameters

Conformance	Search Parameters	Value	Example
-------------	-------------------	-------	---------

SHALL	patient + _revinclude	Patient ID&_revinclude	GET [base]/[Resource]?patient=[id]&_revinclude=Provenance:target
SHALL	id + _revinclude	Resource&_revinclude	GET [base]/[Resource]?_id=[id]&_revinclude=Provenance:target